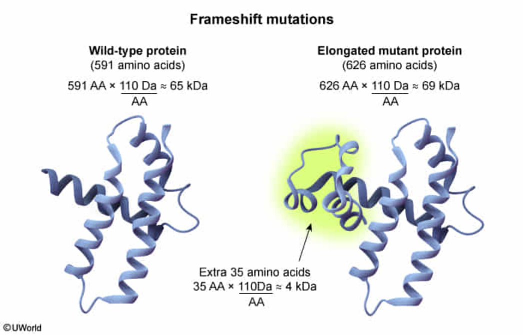
If a frameshift mutation changes the number of amino acids in a protein from 591 to 626, the difference between molecular weights of the wild-type and mutant proteins would be closest to:

- A. 1 kDa [16%]
- B. 2 kDa [16%]
- C. 4 kDa [59%]
- D. 8 kDa [7%]

Correct
60% answered correctly

Explanation:



Frameshift mutations can disrupt the location of stop codons and frequently result in elongated or truncated proteins. The question states that the frameshift mutation increased the length of the protein from 591 to 626 amino acids. The approximate **molecular weight of a protein** can be determined by multiplying the **number of amino acids** by the **average weight of a single amino acid (110 Da)**, so a protein with 591 amino acids (wild type) would have a molecular weight of approximately 65,000 Da, or 65 kDa (110 Da/amino acid × 591 amino acids ≈ 65,000 Da). The molecular weight of the mutant protein is 110 Da/amino acid × 626 amino acids ≈ 69,000 Da, or 69 kDa. The **difference in molecular weight** between 69 kDa and 65 kDa is **4 kDa**.

Alternatively, the answer may be determined by the difference in the number of amino acids (626 − 591 = 35 amino acids) multiplied by 110 (35 amino acids × 110 Da/amino acid = 3850 Da ≈ 4 kDa).

(Choices A, B, and D) Calculating the average molecular weight of amino acids incorrectly could yield these numbers.

Educational objective:
The average molecular weight of an amino acid is 110 Da. The molecular weight of a polypeptide can be estimated by multiplying the number of amino acids in the polypeptide by 110.

Time spent:0 sec
Subject:
Biochemistry

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Amino Acids and Proteins